m=51 deep conductor endpoint discovery

Descriptive report for sealed run m51-deep-scan-run-d2dd22d7

2026-10-13

This is a descriptive discovery report. It records observations only; it does not approve, promote, support, weaken, or reject an ECDLP or weak-curve claim.

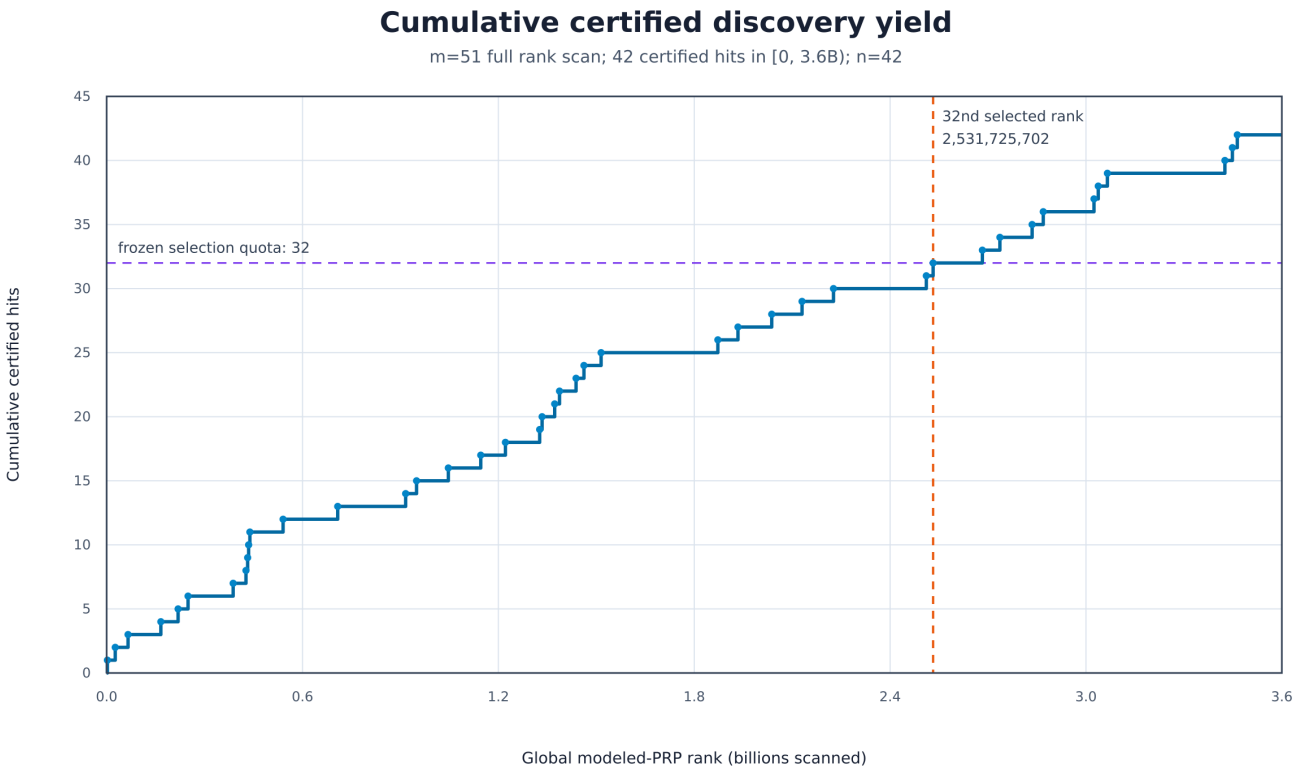
Scope

The run exhausted the frozen global rank interval `[0, 3,600,000,000)` in 256 shards. It searched the ordinary Koblitz isogeny class over `GF(2^51)` with modulus `x^51+x^6+x^3+x+1` and models `E_b: y^2+xy=x^3+b`. The shared curve order was `N=2,251,799,852,369,764`, with `D_K=-7` and `f_pi=32,756,041=271*120871`.

Recorded observations

Quantity	Recorded value
Planned/scanned shards	256 / 256
Sieve hits	42
Certified source-parent pairs	42 / 42
Frozen selected pairs	32
Selection cutoff	2,531,725,702
Admitted failures	0
Blocking failures	0
Passed frozen controls	21 / 21

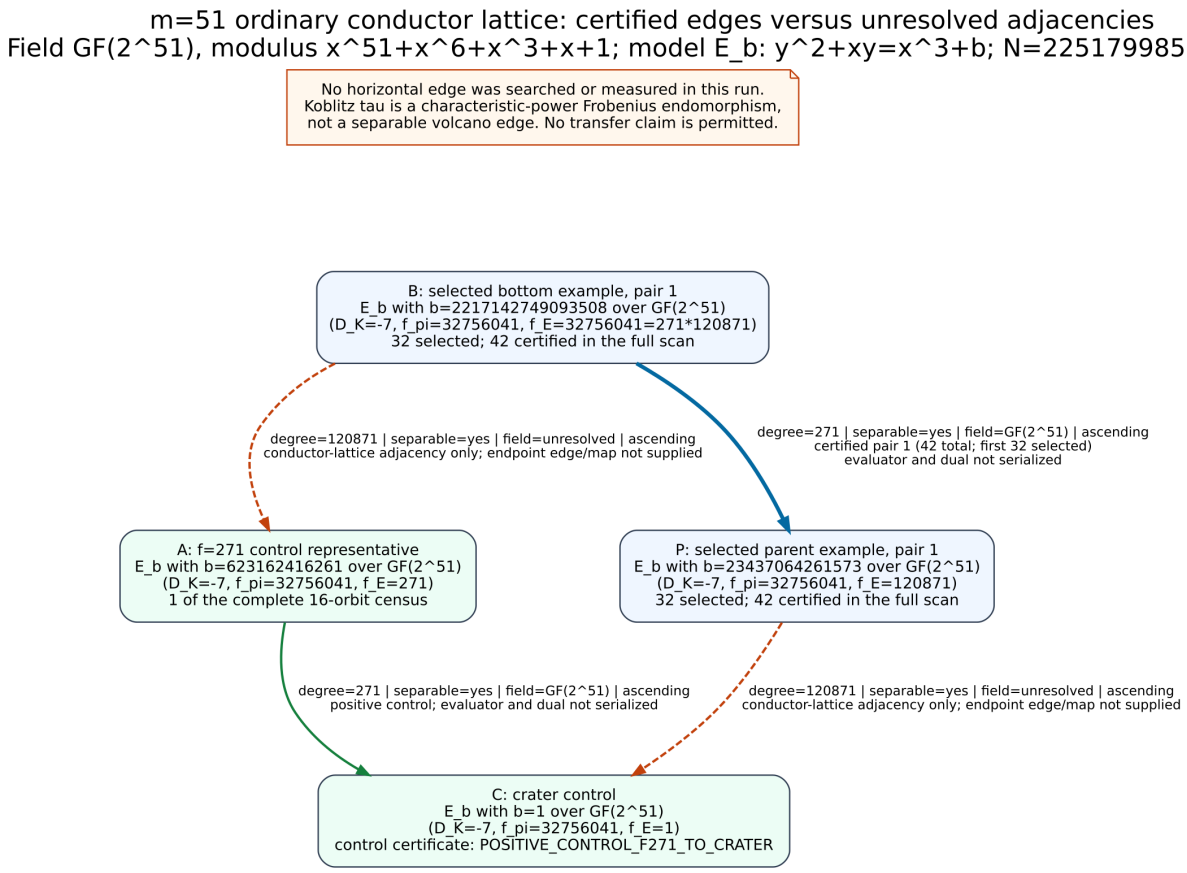
Every certificate records a source/bottom conductor `f_E=32,756,041`, a target/parent conductor `f_E=120,871`, orbit sizes 51, and a separable degree-271 ascending edge over `GF(2^51)`. The first 32 eligible distinct parent orbits were selected in global-rank order; all ten later certified hits remain preserved.



Source: results/hits.csv; sealed run tree m51-deep-scan-run-d2dd22d7. No uncertainty interval was estimated for this descriptive curve.

Conductor-lattice context

The blue edge below is one exact selected bottom-parent example; the same certificate shape holds for all 42 classified hits. The green edge is a frozen positive control. Dashed degree-120871 lines are only conductor-lattice adjacencies: this run supplied neither those endpoint maps nor their field of definition. No horizontal edge was searched or measured.



Artifact and audit summary

The complete deterministic archive is `run1.tar.zst` (2,445,167 bytes), with its SHA-256 and root-manifest SHA-256 bound in `ARCHIVE.md`. All 1,665 listed files passed manifest verification in both the live sealed tree and an extracted archive, and the two trees compared byte-for-byte equal. See `ARCHIVE.md`, `AUDIT.md`, and `ATTEMPTS.md` for the reproducible checks and failure accounting.

Explicit boundary

All 42 certificates record `map_evaluator_serialized=0`, `dual_serialized=0`, and `transfer_claim_permitted=0`. This result therefore supplies certified endpoint-native conductor and edge facts only. It does not supply an executable evaluator or dual, subgroup preservation or scalar recovery, comparative ECDLP measurements, a conductor-caused hardness effect, a weak representative, source-transfer weakness, or weakness of a named curve.